7 Document summarization and RAG with LangChain.js

This chapter covers

- Building a document summarization app
- Implementing retrieval-augmented generation
- Providing grounding support for AI outputs
- Processing documents with advanced techniques

Let's put the tools and techniques we've learned into action with two more sophisticated web apps. First, we'll create a document summarization web application capable of handling two document formats (PDFs and DOCX) implementing advanced semantic chunking strategies and generating meaningful summaries. The application will also show us how to overcome some limitations of document summarization, such as prompt compression and k-means clustering to improve context retention and summary quality.

The second project uses retrieval-augmented generation (RAG) to create a system that dynamically retrieves and synthesizes information based on a list of documents. By combining hybrid search strategies, multistep document processing, and prompt